HyCline Benchmarking Plan

Independent ways to check that every step of the pipeline works

1. Purpose and principles

HyCline's unit tests (469 at the time of writing) check that the code does what it was written to do. Benchmarks answer a different question: **are the results right?** A result is trustworthy when several benchmarks that could fail for *different* reasons all agree. This plan lists, for each step of HyCline, the independent kinds of evidence available and what counts as passing.

Six kinds of benchmark are used throughout. They are deliberately different, because each one has a blind spot the others cover:

Type	What it is	What it catches	Blind spot
A. Exact / hand-computed	Tiny inputs whose answer can be worked out by hand or by brute force (e.g. a 20,001-point likelihood grid).	Arithmetic, indexing, sign and off-by-one errors.	Says nothing about realistic data or statistical behaviour.
B. Invariance / consistency	Transformations that must not change the answer: allele polarity flips, reordering samples or loci, VCF vs beagle at high depth, hard calls vs one-hot genotype likelihoods, missing data vs dropping it, MLE vs flat-prior Bayesian.	Hidden assumptions (the allele-orientation bugs found so far were exactly this kind).	A consistently wrong method passes.
C. Simulation with known truth	Generate data from known parameters and check recovery: bias, error, interval coverage, false-positive and detection rates over many replicates.	Statistical bias, mis-calibrated intervals, sensitivity to depth, missing data and sample size.	Circular if the simulator makes the same assumptions as the model.
D. Other software	Run established tools on identical input and compare (HZAR, gghybrid, bgc/bgchm, ANGSD, bcftools/vcftools/PLINK, ADMIXTURE/NGSadmix, introgress, Clinalyse).	Model or implementation differences; anchors HyCline to accepted practice.	Tools can share the same simplification; differences need explaining, not just measuring.
E. Published empirical results	Reproduce published analyses on their real data (Manacus manakin data bundled with HZAR; the Italian sparrow data bundled with gghybrid; Firneno et al. 2023 Table 1).	Behaviour on real, messy data.	Truth unknown; agreement only shows consistency with the original analysis.
F. Robustness, scale and performance	Edge cases (tiny panels, all-missing loci, monomorphic loci, extreme depth), large inputs, runtime and memory.	Crashes, silent failures, quadratic slow-downs (one was found in estimate-hybrid-index).	Says nothing about accuracy.

Independence rule. A benchmark is only independent if the truth it checks against was not produced by the code under test or by a simulator built on exactly the same model. `benchmarking/simulate_vcf.py` is valuable but shares HyCline's assumptions (independent loci, Hardy-Weinberg proportions, hybrids drawn from the reference panel's exact sample frequencies), so every simulation result should be paired with a type B, D or E benchmark, and with an independent forward simulator (Section 4) where possible.

2. HyCline's steps and functions

The pipeline, and the parts that sit beside it:

Step / function	Commands	Main output
1. AIM filtering	filter-aims	Filtered VCF/beagle + summary (AIM count, missing data, warnings)

Step / function	Commands	Main output
2. Hybrid index	estimate-hybrid-index	Per-individual h with 95% interval
3. Input conversion	generate-geographic-input, generate-genomic-input	Site-level counts or per-individual genotypes/likelihoods (+ parental frequencies for genomic)
4. Geographic clines	fit-geographic (MLE and Bayesian), compare-tails	Centre, width, pmin/pmax, tail parameters, AICc
5. Trait clines	fit-trait, fit-trait-multi	Trait cline parameters (HZAR trait model)
6. Genomic clines	fit-genomic	Per-locus u, v (allele frequency p0 + (p1 - p0) phi(h))
7. Hierarchical clines	fit-hierarchical-geographic, fit-hierarchical-genomic	Pooled per-locus estimates + across-locus means and SDs
8. Coupling / concordance	coupling-geographic, coupling-genomic	Cline variance, theta
9. Genotype-likelihood machinery	beagle input throughout; GL-aware likelihoods	Same outputs, marginalizing genotype uncertainty
10. Whole pipeline	all of the above chained	End-to-end results

3. Benchmarks for each step

Step 1. AIM filtering (filter-aims)

Selects loci whose parental frequencies meet --minor-allele-freq or --allele-freq-diff, and writes a summary with missing-data statistics and warnings. VCF frequencies are sample frequencies; beagle frequencies are pooled maximum-likelihood (EM) estimates.

Benchmark type	What to do	Pass criterion / what it tells you
A. Exact	Small hand-written VCF/beagle files with known frequencies at, just inside and just outside each threshold (including floating-point boundary cases such as 0.57 vs 0.07 for D = 0.5).	Exactly the expected loci kept (already partly in tests/test_filter_aims.py).
B. Invariance	Swap REF/ALT on half the loci; swap --pop1-label/--pop2-label; shuffle sample order.	Identical retained set.
C. Simulation	simulate_vcf.py with --aim-cutoff or --aim-freq-diff, varying panel size (5-50), missing data (0-40%, global and per-individual), and depth for beagle (1-30x). Many seeds per setting.	VCF: retained set equals the simulated AIM set exactly. Beagle: report recall and precision by depth (current: 180-298 of 300 AIMs recovered from 1x to 30x, no false AIMs). Summary missing-data figures equal the simulator's metadata exactly.
D. Other software	Compute per-population allele frequencies with bcftools (+fill-tags with population groups), vcftools --freq --keep, or PLINK --freq --within, and apply the same rule. For beagle, run ANGSD -doMaf per population. Compare per-sample missingness with vcftools --missing-indv or bcftools stats.	VCF: identical AIM set and identical frequencies. Beagle: frequency correlation > 0.99 with ANGSD -doMaf at moderate depth; explain any differences.
E. Empirical	A real dataset with known diagnostic markers (e.g. a published AIM panel) run through filter-aims.	Published diagnostic markers are retained.
F. Robustness	Monomorphic loci, loci missing in a whole population, multiallelic sites/indels, 100k-1M SNPs.	Correct skips and counts; runtime and memory recorded.

Step 2. Hybrid index (estimate-hybrid-index)

Maximizes, per individual, the likelihood of its genotypes given p(h) = (1 - h) p0 + h p1 and Hardy-Weinberg proportions (the gghybrid likelihood without its prior), with a 95% likelihood-profile interval.

Benchmark type	What to do	Pass criterion / what it tells you
A. Exact	Brute-force likelihood grid on small inputs; the worked three-locus example in the Genomic guide (h = 0.576, interval [0.089, 0.964]).	Estimate and interval ends within 1e-4 of the grid (already in tests/test_hybrid_index.py).
B. Invariance	Flip allele polarity at random loci (with p0/p1 flipped); reorder loci; VCF vs beagle at 30x; missing genotypes vs dropping them.	Identical h (to ~1e-8).
C. Simulation	simulate_vcf.py across depth (1-30x), AIM fraction (5-100%), AIM strength, missing data, panel size, and hybrid-index distributions (uniform, beta, fixed). Use the true h in the metadata file (_metadata.tsv).	Mean absolute error and bias by setting; 95% interval coverage 93-97% over enough replicates; no compression toward 0.5. Current: error 0.005-0.016; coverage 0.90-0.97 on 40 hybrids per setting.
C. Simulation (structure)	Pedigree-based hybrids: F1, F2 and backcrosses built from parental genotypes (not independent draws per locus).	F1s at h ~ 0.5, backcrosses at ~0.25/0.75; checks the independence assumption the simulator currently shares with the model.
D. Other software	gghybrid esth with a uniform Beta(1,1) prior on the same loci and parental samples; bgc / bgchm hybrid index; ADMIXTURE (or NGSadmix for genotype likelihoods) with K = 2; introgress.	gghybrid posterior mode should match closely (same likelihood); ADMIXTURE/NGSadmix Q should correlate > 0.98 (different model: admixture proportion, not hybrid index).
E. Empirical	Italian sparrow data bundled with gghybrid (Hermansen et al. 2014); compare with gghybrid's published hybrid indices.	Close agreement; parental samples near 0/1.
F. Robustness / scale	Individuals with very few called loci; 100k-1M loci; low depth.	Clear errors for unusable individuals; record runtime (currently ~49 s for 100k SNPs x 160 samples). Note: with many weakly differentiated loci intervals become overconfident (coverage 57% on 100k SNPs, 94% on its AIMs) - keep this as a standing benchmark.

Step 3. Input conversion (generate-geographic-input, generate-genomic-input)

Converts VCF/beagle + metadata into HyCline CSVs. The genomic converter also writes each locus's parental frequencies (p0/p1, clipped to [0.001, 0.999]) and leaves reference individuals out by default.

Benchmark type	What to do	Pass criterion / what it tells you
A. Exact	Small files where site aggregates, genotype frequencies, log genotype likelihoods and p0/p1 can be computed by hand.	Exact equality.
B. Invariance	Reorder samples in the VCF vs the metadata; gzip vs plain; extra FORMAT fields.	Identical CSV (up to row order).
D. Other software	bcftools query -f '[%GT\t]' and a short script to aggregate per site; compare p0/p1 with bcftools/PLINK per-population frequencies (VCF) or ANGSD -doMaf (beagle).	Exact agreement for VCF; close agreement for beagle EM frequencies.
F. Robustness	Samples missing from metadata, loci with no data, equal parental frequencies, parental reference labels given or omitted.	Correct drops and counts, clear messages.

Step 4. Geographic clines (fit-geographic, compare-tails)

Fits sigmoid clines (centre, width, pmin/pmax), with optional HZAR tail models, by MLE or Bayesian NUTS, from hard counts or genotype likelihoods.

Benchmark type	What to do	Pass criterion / what it tells you
A. Exact	Likelihood values at fixed parameters vs hand computation; HZAR's cline equations evaluated directly.	Equality to rounding.

Benchmark type	What to do	Pass criterion / what it tells you
B. Consistency	MLE vs NUTS with flat priors on well-informed data; hard counts vs one-hot genotype likelihoods; site order.	Agreement within uncertainty (already documented: MLE matches HZAR on the manakin Ada locus).
C. Simulation	Needs a transect simulator (Section 4): sites along a line, per-site hybrid index from a known geographic cline, genotypes per locus with known centre/width, optional tails. Vary number of sites, individuals per site, depth.	Bias and RMSE of centre and width; 95% interval coverage; compare-tails picks the generating tail model at a known rate.
D. Other software	HZAR on identical data (all model variants: fixed/free ends, tails); Clinalyse for T-clines once implemented.	Parameters and log-likelihoods match HZAR's MLE; AICc ranks agree.
E. Empirical	Manacus manakin data bundled with HZAR: all 7 molecular loci, reproducing Derryberry et al. (2014) Tables 3-4.	Already done for Ada (centre 171.0 vs 170.9 km); extend to all loci and tail models.
F. Robustness	Few sites, sites only on one side, saturated frequencies, tail models under Bayesian fitting (known convergence difficulty).	Clear warnings; convergence diagnostics flag problems.

Step 5. Trait clines (fit-trait, fit-trait-multi)

HZAR's quantitative-trait cline model, including tail models, verified against HZAR's R source.

Benchmark type	What to do	Pass criterion / what it tells you
A. Exact	Gaussian likelihood at fixed parameters vs HZAR's R functions evaluated on the same input.	Equality to rounding.
C. Simulation	Simulate trait values per individual from known cline mean/variance profiles (including unequal parental variances and tails).	Recovery of centre, width and variances; interval coverage.
D. Other software	HZAR trait models on identical data.	Matching estimates and log-likelihoods.
E. Empirical	Trait data analysed in Derryberry et al. (2014) or other published trait-cline studies.	Reproduce the published parameters.

Step 6. Genomic clines (fit-genomic)

Per-locus logit-logistic cline: allele frequency p0 + (p1 - p0) phi(h), logit(phi) = u + v logit(h), by MLE from hard counts or genotype likelihoods.

Benchmark type	What to do	Pass criterion / what it tells you
A. Exact	Likelihood at fixed (u, v) vs hand computation; allele-frequency form with p0 = 0, p1 = 1 equals phi.	Equality (in tests).
B. Invariance	Flip polarity at a locus (swap counts and p0/p1); include vs exclude parental references.	Same u, v (reference exclusion should change little on large samples).
C. Simulation, neutral	simulate_vcf.py (every locus neutral, u = 0, v = 1) through the full pipeline.	Median u ~ 0, v ~ 1 for every locus class; false-positive rate ~5% per parameter. Current: median u 0.02, v 1.07; u 1.5%, v 6.5% (VCF); v 9.2% at 5x beagle - investigate.
C. Simulation, selection	Needs a simulator extension: per-locus (u, v) for hybrids (barrier loci v > 1, permeable v < 1, biased u != 0).	Recovery of u, v; detection power vs false-positive rate; ranking of outliers.
D. Other software	gghybrid ggcline (same model; note gghybrid's u has the opposite sign, and it reports centre with logit(centre) = u/v); bgc (different parameterisation - compare outlier sets and ranks); introgress.	Close agreement with gghybrid; overlapping outlier sets with bgc.
E. Empirical	Italian sparrow data with gghybrid and BGC output (77-locus comparison already done for BGC); rerun with parental frequencies.	Agreement at least as good as before.

Step 7. Hierarchical clines (fit-hierarchical-geographic / -genomic)

Partial pooling of per-locus parameters with NUTS; hard counts or genotype likelihoods.

Benchmark type	What to do	Pass criterion / what it tells you
C. Simulation	Simulate per-locus parameters from known across-locus distributions (known mu and sigma for u and log v, or centre and log width), including all-neutral (sigma = 0).	Recovery of mu and sigma; shrinkage strongest for data-poor loci; sigma near 0 when there is no real variation (current short runs gave 0.2-0.3 - check with longer chains).
B. Consistency	With weak pooling (large sigma priors), per-locus posteriors approach independent MLE fits.	Agreement within uncertainty.
D. Other software	bgchm (Gompert, DeRaad & Buerkle 2024) hierarchical genomic clines on identical data.	Similar across-locus SDs and per-locus estimates.
F. Robustness	Convergence diagnostics (R-hat, ESS, divergences) across seeds; runtime vs number of loci.	Diagnostics clean on standard settings; record time.

Step 8. Coupling / concordance (coupling-geographic, coupling-genomic)

Across-locus cline variance and Firneno et al. (2023)'s coupling coefficient theta.

Benchmark type	What to do	Pass criterion / what it tells you
E. Published	Reproduce Firneno et al. (2023) Table 1 (already documented).	Values match the published table.
C. Simulation	Simulate cline sets with known across-locus variance (and, where possible, from forward simulations with known selection and dispersal - Section 4).	Recovered variance and theta track the simulated values; record bias.

Step 9. Genotype-likelihood machinery (beagle input throughout)

Marginalizes genotype uncertainty instead of using hard calls.

Benchmark type	What to do	Pass criterion / what it tells you
B. Consistency	One-hot likelihoods from hard calls give exactly the hard-count likelihood; high-depth beagle gives the same results as true-genotype VCF.	Equality / near-equality.
C. Simulation	simulate_vcf.py --format-beagle with --vcf-genotypes called gives matched hard-call and likelihood versions of one dataset across depth, dispersion and per-individual depth.	Likelihood path stays accurate as depth falls; hard calls degrade (heterozygotes called correctly 25% at 1x, 100% at 30x).
D. Realistic upstream	Simulate reads (wgsim or ART) from genotypes, align (bwa), and produce beagle files with ANGSD -doGlf 2 instead of HyCline's own likelihood simulator.	Checks HyCline against real genotype-likelihood output rather than its own idealised model.

Step 10. Whole pipeline

filter-aims -> estimate-hybrid-index -> generate-*-input -> fitting -> coupling.

Benchmark type	What to do	Pass criterion / what it tells you
C. Simulation	End-to-end runs on simulated zones (neutral, and with barrier/permeable loci), both VCF and beagle, at several depths and panel sizes.	Final per-locus parameters and outlier calls match the truth at the rates established step by step.
C. Forward simulation	SLiM or msprime hybrid zones with selection, linkage and geography (Section 4).	Qualitative truth: barrier loci steeper, linked loci concordant, neutral loci near the genome-wide average.
E. Empirical	Manakin (geographic) and Italian sparrow (genomic) end to end.	Reproduces the published conclusions.

4. Simulators: what exists and what to add

`benchmarking/simulate_vcf.py` currently provides: pop1/pop2/popH samples; AIMs by minor-allele-frequency cutoff or frequency difference; exact global and per-individual missing data; hybrid-index distributions; beagle genotype likelihoods from simulated reads (depth, dispersion, per-individual depth, sequencing error, called-genotype VCFs); population-frequency hybrids and reference panels (`--hybrid-freqs population`); and non-neutral genomic clines (barrier, permeable and biased classes among the AIMs, plus random across-locus variation in u and log v). Its truth files give AIM status, panel and source frequencies, cline class, u and v, and true hybrid indices; every run also writes a `_params.tsv` record of its options, command line, git commit and realized counts (metadata, truth and params files are tab-delimited).

Extension	Why it is needed	Benchmarks it enables
Done: hybrids drawn from population frequencies, with the reference panel a sample of them (--hybrid-freqs population)	In the default panel mode hybrids use the panel's exact sample frequencies, so VCF benchmarks give HyCline error-free parental frequencies.	Realistic accuracy for filter-aims, hybrid index and clines; the cost of small panels. First results in Section 6.
Done: non-neutral genomic clines (--barrier/--permeable/--biased-percent, --cline-u-sd, --cline-log-v-sd)	Every simulated locus used to be neutral.	Power, outlier detection, recovery of u and v (Step 6), hierarchical sigma (Step 7).
Geographic transects (sites, distances, clines with tails)	No geography at present.	Steps 4 and 7 (geographic), compare-tails, coupling-geographic.
Quantitative traits	No trait simulation.	Step 5.
Pedigree hybrid classes (F1, F2, backcrosses)	Independent per-locus draws can't produce F1-like genotype structure.	Hybrid-index accuracy for real hybrid classes; comparison with NewHybrids.
Linkage between loci	Loci are independent now.	Effect of LD on intervals and on the many-loci overconfidence seen for hybrid index.
Read-level pipeline (wgsim/ART + bwa + ANGSD)	HyCline's likelihood simulator matches HyCline's model by construction.	An independent test of every beagle path (Step 9).
Forward simulations (SLiM; msprime for neutral backgrounds)	Parametric simulators share the fitted model's assumptions.	Independent truth from evolutionary processes (selection against hybrids, dispersal, tension zones) - the strongest test of whether the clines mean what the documentation says they mean.

5. Metrics and acceptance criteria

Metric	Use for	Target / how to read it
Bias (mean estimate - truth) and RMSE	All parameters with simulated truth	Bias within Monte Carlo error; RMSE shrinking as data increase.
95% interval coverage	Hybrid-index intervals, cline-parameter CIs, Bayesian credible intervals	93-97% given replicates; with 100 replicates the Monte Carlo error is about +/-2.2 points, so use several hundred where possible.
False-positive rate	Neutral loci flagged as non-neutral; non-AIMs kept as AIMs	Close to the nominal rate per test (5% per parameter); remember u-or-v flags two tests (~10%).
Power / recall and precision	Detecting AIMs, barrier/permeable loci	Reported as curves vs depth, sample size and effect size.
Agreement with other software	Every type D comparison	Exact where the model is identical; otherwise correlation, Bland-Altman limits, and an explanation of systematic differences.
Convergence diagnostics	All NUTS fits	R-hat < 1.01, bulk/tail ESS > 400, no divergences.
Runtime and peak memory	Every command at several sizes	Recorded per release; no unexplained regressions.

6. Open issues these benchmarks should track

- Wald intervals for genomic-cline v are narrower than nominal at low depth (9.2% vs 5% at 5x with 60 hybrids; 7.7% even with true frequencies and hybrid index). Compare with profile-likelihood intervals.
- Parental frequencies are treated as exact everywhere (as in gghybrid). With many weakly differentiated loci this makes hybrid-index intervals overconfident (57% coverage on 100k SNPs). Benchmark with the population-frequency simulator extension.
- Hierarchical across-locus SDs were 0.2-0.3 on all-neutral data in short runs; check with longer chains.
- First population-frequency results (15 + 15 references, 100 hybrids, 761 AIMs): with every locus neutral, hybrid-index error rose from 0.011 (panel mode) to 0.020 and 95% interval coverage fell from 0.93 to 0.68 -- the parental frequencies' sampling error, which the estimator treats as exact. filter-aims missed 13-20% of true AIMs (those near the threshold) and, with 5 references per population, 6% of what it kept were not AIMs.
- With 45% of AIMs non-neutral (15% each barrier v = 4, permeable v = 0.25, biased |u| = 2), the hybrid index -- whose model assumes neutral loci -- degraded (error 0.026-0.037, coverage 0.54-0.58), and fitted clines were pulled toward neutral (barrier v 3.3-3.9, permeable 0.33-0.37). Given the true hybrid index and source frequencies the same fits were accurate (barrier 4.2-4.3, permeable 0.25-0.26, neutral 1.02-1.04), so the cline fitting itself is sound and the attenuation comes from the estimated inputs. Benchmark realistic outlier fractions next, and consider excluding loci flagged as outliers from the hybrid-index set in a second pass.
- The genomic pipeline without parental frequencies (legacy mode) still pins reversed loci at the v bound; keep the regression benchmark that documents it.
- tests/test_mcmc_numpyro.py::test_diagnostics_check_runs_and_flags_nothing_on_a_clean_fit fails on main in the current environment (R-hat 1.0136 vs 1.01); unrelated to the benchmarks but should be resolved.

7. How to organise and run the benchmarks

- Keep benchmarks separate from unit tests: `benchmarking/` holds one script per benchmark, each writing a results table (CSV) and a short summary; unit tests keep checking code behaviour quickly.
- Fix and record seeds, HyCline version (git commit), and versions of every external tool (HZAR, gghybrid, bgc/bgchm, ANGSD, bcftools, PLINK, ADMIXTURE, SLiM).
- Run replicates (100+ per setting for coverage and false-positive rates) and report Monte Carlo error.
- Re-run the full benchmark set before releases and after changes to models, likelihoods or converters; compare against the previous results table.
- When a benchmark fails, follow the project's rule: find the root cause, add a regression test built from the data that exposed it, and record the finding in BACKEND_*.md.

8. Suggested order

Priority	Work	Reason
1	**Done:** population-frequency hybrids and non-neutral genomic clines in simulate_vcf.py (first results in Section 6)	Removes the main circularity and enables power/outlier benchmarks for the genomic pipeline.
2	gghybrid comparison (esth + ggcline) on simulated and Italian sparrow data	Same model, so differences point straight at bugs.
3	bcftools/PLINK and ANGSD -doMaf comparisons for filter-aims and parental frequencies	Cheap and exact for VCF.
4	Transect simulator + HZAR comparison on all manakin loci and tail models	Covers the geographic and trait half of HyCline.
5	Read-level beagle pipeline (wgsim/ART + bwa + ANGSD)	Independent check of every genotype-likelihood path.
6	Pedigree hybrid classes, linkage, SLiM forward simulations	Tests assumptions shared by the simulator and the models.
7	bgc/bgchm and ADMIXTURE/NGSadmix comparisons; scale and performance suite	Broader cross-software and practical limits.